



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: PROJECT EVALUATION (24CS2235F)

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TITLE OF THE PROJECT	Mini Compiler for a Custom Programming Language	
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Abstract

The **Mini Compiler for a Custom Programming Language** is a compiler design project that implements the major phases of a compiler for a simple, user-defined programming language. The system accepts source code written in the custom language and transforms it into optimized target code through a systematic compilation pipeline. The process begins with **lexical analysis**, where the source program is converted into tokens using regular expressions and **Deterministic Finite Automata (DFA)**. These tokens are then processed by a parser based on a **Context-Free Grammar (CFG)** to verify the syntactic structure of the program and generate an **Abstract Syntax Tree (AST)**. The AST is used for syntax-directed translation and **Three Address Code (TAC)** generation, which provides an intermediate representation of the source program. The generated intermediate code is optimized using techniques such as **constant folding, constant propagation, common subexpression elimination, and dead code elimination** to reduce unnecessary computations and improve efficiency. Finally, the optimized intermediate code is converted into simple **target or machine-like code** with basic register allocation. The proposed compiler demonstrates the complete compilation process from source code to target code and provides practical implementation of important Compiler Design concepts, including lexical analysis, finite automata, parsing, syntax-directed translation, abstract syntax trees, intermediate code generation, code optimization, and code generation. The project aims to provide a clear and practical understanding of how modern compilers transform high-level programs into efficient low-level representations.

Signature of the Guide

HOD